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Mission Field: Cyber Security

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Day 1: Cybersecurity Basics (CIA, Threats, Risk)

The CIA Triad:

The CIA Triad is the foundational model for information security, consisting of three core principles:

Confidentiality:

Ensures that sensitive information is accessible only to authorized individuals.

Techniques: Encryption, access controls, authentication, data classification

Example: Using AES-256 encryption for data at rest and TLS 1.3 for data in transit

Integrity:

Maintains the accuracy and trustworthiness of data over its entire lifecycle.

Techniques: Hashing (SHA-256), digital signatures, version control, checksums

Example: A bank verifying that transaction amounts haven't been altered during processing

Availability:

Ensures that systems and data are accessible when needed by authorized users.

Techniques: Redundancy, backups, load balancing, disaster recovery planning, DDoS protection

Example: Maintaining 99.99% uptime SLA for critical healthcare systems.

Common Cyber Threats:

Category	Description	Examples
Malware	Malicious software designed to damage or gain unauthorized access	Ransomware, Trojans, worms, spyware

Category	Description	Examples
Phishing	Social engineering attacks via deceptive communications	Spear phishing, whaling, vishing
Network Attacks	Exploiting network vulnerabilities	Man-in-the-middle, DDoS, packet sniffing
Insider Threats	Malicious or negligent actions by internal personnel	Data theft, privilege abuse, accidental leaks
Advanced Persistent Threats (APTs)	Sophisticated, long-term targeted attacks	Nation-state actors, organized cybercrime
Zero-Day Exploits	Attacks targeting unknown vulnerabilities	Unpatched software flaws

Risks Risk Assessment Process:

1. Identify Assets → 2. Identify Threats → 3. Identify Vulnerabilities



4. Assess Impact & Likelihood → 5. Calculate Risk Level → 6. Implement Controls

Thank you